Important connective tissue disorders: (Heritable types)

1. **Ehler's -Danlos syndrome** – Defect in the synthesis of collagen in the joints, heart valves, organ walls and arterial walls.
2. **Stickler syndrome** – Affects collagen and results in facial abnormalities.
3. **Rhabdomyosarcoma** – Life threatening soft tissue tumour of head, neck and urinogenital tract.

Autoimmune connective tissue disorders

1. **Rheumatoid arthritis:** The immune cells attack and inflame the membranes around the joints. It can also affect heart, lungs and eyes.
2. **Sjogren's syndrome:** Progressive inability to secrete saliva and tears.

3.4 Muscle Tissue

Each muscle is made of many long, cylindrical fibres arranged in parallel arrays. These fibres are composed of numerous fine fibrils, called **myofibrils**. Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their uncontracted state in a coordinated fashion. In general muscles play an active role in all the movements of the body.

Muscles are of three types, skeletal, smooth and cardiac. **Skeletal muscle tissue** is closely attached to skeletal bones. In a typical muscle such as the biceps, the striated (striped) skeletal muscle fibres are bundled together in a parallel fashion. A sheath of tough connective tissue encloses several bundles of muscle fibres (You will learn more about this in Chapter 9).

The **smooth muscle** fibres taper at both ends (fusiform) and do not show striations (Figure 3.7). Cell junctions hold them together and they are bundled together in a connective tissue sheath. The walls of internal organs such as the blood vessels, stomach and intestine contain this type of muscle tissue. Smooth muscles are 'involuntary' as their functions cannot be directly controlled. Unlike the smooth muscles, skeletal muscles can be controlled by merely thinking.

Cardiac muscle tissue is a contractile tissue present only in the heart. Cell junctions fuse the plasma membranes of cardiac muscle cells and make them stick together. Communication junctions (intercalated discs) at some fusion points allow the cells to contract as a unit, i.e., when one cell receives a signal to contract, its neighbours are also stimulated to contract.

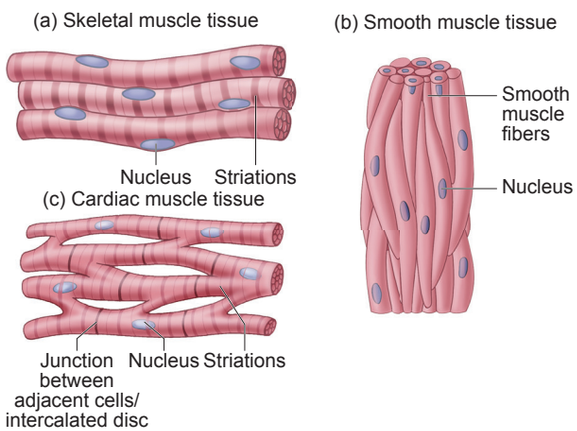


Figure 3.7 Muscle tissues

Palmaris muscle:

This long narrow muscle runs from the elbow to the wrist and is important for hanging and climbing in primates, is missing in 11% of humans today.

3.5 Neural Tissue

Nervous tissue exerts the greatest control over the body's responsiveness to changing

TN_GOVT_XI_Bio Zoology_Ch03.indd 44 09-01-2020 5.26.10 PM